Data Analysis & Probability

ClipSAT Printable Practice Suite (clipsat.org) • Topic 7.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Reading bar charts**

The bar chart shows books sold by genre in one week. Find the total books sold and the modal (most common) genre.

Total $= 45 + 30 + 20 + 15 = 110$ books. The modal genre is Fiction, with 45 books.

2 **[Foundational]** • **Target: Reading bar charts**

The bar chart shows monthly rainfall in mm. Find the mean monthly rainfall.

Mean $= \frac{60 + 45 + 80 + 55}{4} = \frac{240}{4} = 60$ mm.

3 **[Foundational]** • **Target: Mean, median, mode and range**

Find the mean of $6, 9, 12, 15, 18$.

Mean $= \frac{6 + 9 + 12 + 15 + 18}{5} = \frac{60}{5} = 12$.

4 **[Foundational]** • **Target: Mean, median, mode and range**

Ages of 7 students: $13, 15, 13, 16, 14, 13, 17$. Find the mean, median, and mode.

Ordered: $13, 13, 13, 14, 15, 16, 17$. Mean $= \frac{101}{7} \approx 14.43$. Median (4th value) $= 14$. Mode $= 13$ (appears three times).

5 **[Foundational]** • **Target: Mean, median, mode and range**

Find the range and mode of the data set $8, 3, 8, 15, 8, 22, 3$.

Range $= 22 - 3 = 19$. Mode $= 8$ (appears three times).

6 **[Foundational]** • **Target: Mean, median, mode and range**

A data set has mean 18 and 6 values. Find the sum of the values.

Sum $= 18 \times 6 = 108$.

7 **[Foundational]** • **Target: Mean, median, mode and range**

The data set $10, 12, 11, 13, 50$ contains an outlier. Find the mean before and after removing 50.

Before: $\frac{10 + 12 + 11 + 13 + 50}{5} = \frac{96}{5} = 19.2$. After removing 50: $\frac{46}{4} = 11.5$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Mean, median, mode and range

A data set of 5 values has mean 20. Four of the values are $15, 18, 22, 25$. Find the fifth value.

Total sum $= 20 \times 5 = 100$. Known sum $= 15 + 18 + 22 + 25 = 80$. Fifth value $= 100 - 80 = 20$.

9 [Intermediate] • Target: Reading tables

A survey of 50 people's favorite fruit: Apple 15, Banana 12, Orange 10, Mango 13. What fraction of people chose banana?

$\frac{12}{50} = \frac{6}{25}$.

10 [Intermediate] • Target: Reading tables

Using the same survey (Apple 15, Banana 12, Orange 10, Mango 13, out of 50), what percent chose either apple or mango?

$15 + 13 = 28$ people. $\frac{28}{50} \times 100\% = 56\%$.

11 [Intermediate] • Target: Reading tables

A table shows quiz scores: score 5 (3 students), score 7 (5), score 8 (8), score 9 (4), score 10 (2). Find the mode score and the total number of students.

Mode $= 8$ (8 students, the highest frequency). Total students $= 3 + 5 + 8 + 4 + 2 = 22$.

12 [Intermediate] • Target: Reading tables

Using the same quiz data, find the mean score to two decimal places.

Sum $= 5(3) + 7(5) + 8(8) + 9(4) + 10(2) = 15 + 35 + 64 + 36 + 20 = 170$. Mean $= \frac{170}{22} \approx 7.73$.

13 [Intermediate] • Target: Reading bar charts

Using the same book-sales data, what percent of total books sold were mystery novels, to the nearest percent?

$\frac{30}{110} \times 100\% \approx 27\%$.

14 [Intermediate] • Target: Reading bar charts

Using the same rainfall data (Jan 60, Feb 45, Mar 80, Apr 55 mm), find the percent decrease from March to April.

Decrease $= 80 - 55 = 25$ mm. $\frac{25}{80} \times 100\% = 31.25\%$.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Basic probability • Cross-Lesson: 8.01 Word Problems & Reasoning

A bag holds 4 red and 6 blue marbles. Find $P(\text{red})$.

$P(\text{red}) = \frac{4}{10} = \frac{2}{5}$.

16 [Challenging] • Target: Basic probability • Cross-Lesson: 1.01 Arithmetic & Number Sense

A fair six-sided die is rolled. Find $P(\text{a number greater than } 4)$.

Favorable outcomes $\{5, 6\}$: $P = \frac{2}{6} = \frac{1}{3}$.

17 [Challenging] • Target: Basic probability • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

A card is drawn from a standard deck of 52 cards. Find $P(\text{a heart})$.

$P = \frac{13}{52} = \frac{1}{4}$.

18 [Challenging] • Target: Basic probability • Cross-Lesson: 3.01 Algebra Essentials

A spinner has 8 equal sections numbered 1 to 8. Find $P(\text{an even number})$.

Even outcomes $\{2, 4, 6, 8\}$: $P = \frac{4}{8} = \frac{1}{2}$.

19 [Challenging] • Target: Combined and complementary events • Cross-Lesson: 4.01 Geometry & Measurement

A bag holds 3 red, 5 blue and 2 green marbles. Find $P(\text{not red})$.

$P(\text{red}) = \frac{3}{10}$, so $P(\text{not red}) = 1 - \frac{3}{10} = \frac{7}{10}$.

20 [Challenging] • Target: Combined and complementary events • Cross-Lesson: 5.01 Quantitative Comparisons

Two fair coins are tossed. Find $P(\text{both land heads})$.

$P = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: MCQ — Mean, median, mode and range

The bar chart shows quiz scores. Find the total number of students, and the modal score.

a. 20 **students; mode** $= 8$ ✓

b. 20 students; mode $= 6$

c. 20 students; mode $= 9$

d. 18 students; mode $= 8$

Total $= 2 + 4 + 6 + 5 + 3 = 20$; the score 8 has the highest frequency (6 students).

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22 [Foundational] • Target: MCQ — Reading bar charts

The bar chart shows books sold by genre in one week. Find the total books sold, and the modal genre.

a. 120; Romance

b. 100; Fiction

c. 120; Mystery

d. 120; **Fiction** ✓

Total $= 45 + 30 + 25 + 20 = 120$; Fiction had the most sales (45).

23 [Foundational] • Target: MCQ — Reading bar charts

The bar chart shows quarterly sales for four stores, in thousands of riyals. Find the range, and which store had the highest sales.

a. 55; Store B

b. 55; **Store D** ✓

c. 140; Store D

d. 35; Store D

Range $= 140 - 85 = 55$; Store D had the highest sales (140).

24 [Foundational] • Target: MCQ — Reading bar charts

The bar chart shows hours studied each weekday. Find the total hours, and the mean hours per day.

a. 12; 2

b. 12; 2.4 ✓

c. 12; 3

d. 10; 2

Total $= 2 + 3 + 1 + 4 + 2 = 12$; mean $= 12 \div 5 = 2.4$.

25 [Foundational] • Target: MCQ — Reading bar charts

The bar chart shows the number of defective items found for four products in a quality check. Find the total defects, and the range.

a. 40; 10 ✓

b. 40; 7

c. 35; 10

d. 40; 15

Total $= 12 + 8 + 15 + 5 = 40$; range $= 15 - 5 = 10$.

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26 [Foundational] • Target: MCQ — Word problems

The bar chart shows a company's quarterly sales, in thousands of riyals. Find the total annual sales, and the mean quarterly sales.

a. 1,560; 450

b. 1,560; 390 ✓

c. 1,560; 400

d. 1,200; 390

Total $= 320 + 410 + 380 + 450 = 1{,}560$; mean $= 1{,}560 \div 4 = 390$.

27 [Foundational] • Target: MCQ — Mean, median, mode and range

Find the mean of $8, 11, 14, 17, 20$.

a. 12

b. 17

c. 70

d. 14 ✓

Sum $= 70$; mean $= 70 \div 5 = 14$.

28 [Foundational] • Target: MCQ — Mean, median, mode and range

A data set of 6 values has mean 25. Five of the values are $20, 22, 28, 30, 32$. Find the sixth value.

a. 18 ✓

b. 28

c. 8

d. 150

Total needed $= 6 \times 25 = 150$; sum of five $= 132$; sixth $= 150 - 132 = 18$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: MCQ — Reading tables

A table shows the number of pets owned by 25 families: 0 pets (6 families), 1 pet (10), 2 pets (7), 3 pets (2). Find the median number of pets.

a. 0

b. 1.2

c. 2

d. 1 ✓

With 25 values sorted, the median is the 13th value, which falls in the '1 pet' group.

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30 [Intermediate] • Target: MCQ — Basic probability

A card is drawn from a standard deck of 52 cards. Find $P(\text{a king})$.

a. $\frac{1}{4}$

b. $\frac{1}{52}$

c. $\frac{1}{13}$ ✓

d. $\frac{4}{13}$

4 kings out of 52: $\frac{4}{52} = \frac{1}{13}$.

31 [Intermediate] • Target: MCQ — Combined and complementary events

Two fair coins are tossed. Find $P(\text{at least one tail})$.

a. $\frac{3}{4}$ ✓

b. $\frac{1}{3}$

c. $\frac{1}{4}$

d. $\frac{1}{2}$

$P(\text{no tails}) = \frac{1}{4}$, so $P(\text{at least one}) = 1 - \frac{1}{4} = \frac{3}{4}$.

32 [Intermediate] • Target: MCQ — Weighted averages

A class of 12 students has an average score of 78. A 13th student joins with a score of 91. Find the new average.

a. 84.50

b. 78.00

c. 91.00

d. 79.00 ✓

New total $= 12(78) + 91 = 1{,}027$; new average $= 1{,}027 \div 13 = 79.00$.

33 [Intermediate] • Target: MCQ — Weighted averages

Two classes combine: Class A (10 students, mean 60) and Class B (n students, mean 90), giving a combined mean of 78 for all students. Find n.

a. 12

b. 15 ✓

c. 10

d. 20

$\frac{600 + 90n}{10 + n} = 78 \Rightarrow 600 + 90n = 780 + 78n \Rightarrow 12n = 180 \Rightarrow n = 15$.

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34 [Intermediate] • Target: MCQ — Comparing data sets

Two data sets both have median 20. Set P: $10, 15, 20, 25, 30$. Set Q: $18, 19, 20, 21, 22$. Which set is more spread out?

a. **Set P ✓**

b. Cannot be determined

c. Set Q

d. They are equally spread out

Range P $= 20$; range Q $= 4$. Set P is more spread out.

35 [Intermediate] • Target: MCQ — Word problems

A survey of 150 shoppers found 90 prefer online shopping and the rest prefer in-store. Find the probability, as a fraction in simplest form, that a randomly chosen shopper prefers in-store.

a. $\frac{2}{5}$ ✓

b. $\frac{90}{150}$

c. $\frac{3}{5}$

d. $\frac{3}{50}$

In-store $= 60$ out of 150, which simplifies to $\frac{2}{5}$.

36 [Intermediate] • Target: MCQ — Word problems

Find the mean of $24, 23, 48, 18, 12$.

a. 30

b. 25 ✓

c. 23

d. 27

Mean $= \frac{24+23+48+18+12}{5} = \frac{125}{5} = 25$.

37 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the mean of $50, 50, 43, 21, 21$.

a. 39

b. 42

c. 35

d. 37 ✓

Mean $= \frac{50+50+43+21+21}{5} = \frac{185}{5} = 37$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 3.01 Algebra Essentials

Find the mean of $55, 7, 53, 41, 49$.

a. 39

b. 46

c. 41 ✓

d. 43

Mean $= \frac{55+7+53+41+49}{5} = \frac{205}{5} = 41$.

39 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Geometry & Measurement

A bag holds 8 red and 9 blue marbles. Find $P(\text{red})$.

a. $\frac{8}{17}$ ✓

b. $\frac{10}{17}$

c. $\frac{9}{17}$

d. $\frac{10}{19}$

$P(\text{red}) = \frac{8}{17}$.

40 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 5.01 Quantitative Comparisons

A bag holds 11 red and 5 blue marbles. Find $P(\text{red})$.

a. $\frac{5}{16}$

b. $\frac{3}{4}$

c. $\frac{13}{18}$

d. $\frac{11}{16}$ ✓

$P(\text{red}) = \frac{11}{16}$.

41 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 6.01 Sequences & Patterns

A bag holds 11 red and 9 blue marbles. Find $P(\text{red})$.

a. $\frac{13}{22}$

b. $\frac{9}{20}$

c. $\frac{11}{20}$ ✓

d. $\frac{3}{5}$

$P(\text{red}) = \frac{11}{20}$.

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 8.01 Word Problems & Reasoning

Class A has 26 students with mean score 65; Class B has 22 students with mean score 89. Find the overall mean of all 48 students.

a. 78

b. 76 ✓

c. 77

d. 74

Total sum $= 26(65) + 22(89) = 1690 + 1958 = 3648$. Overall mean $= \frac{3648}{48} = 76$.

43 [Challenging] • Target: Mean, median, mode and range • Cross-Lesson: 1.01 Arithmetic & Number Sense

Event X has probability $\frac{1}{7}$ and independent event Y has probability $\frac{4}{5}$. Find $P(X \text{ and } Y)$.

a. $\frac{1}{70}$

b. $\frac{4}{35}$ ✓

c. $\frac{33}{35}$

d. $\frac{3}{14}$

$P(X \text{ and } Y) = \frac{1}{7} \times \frac{4}{5} = \frac{4}{35}$.

44 [Challenging] • Target: Reading tables • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

Find the mean and median of $27, 24, 37, 38, 14$.

a. $27, 28$

b. $30, 27$

c. $28, 29$

d. $28, 27$ ✓

Ordered: $14, 24, 27, 37, 38$. Mean $= \frac{14+24+27+37+38}{5} = \frac{140}{5} = 28$. Median (3rd value) $= 27$.

45 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Algebra Essentials

A bag holds 6 red and 9 blue marbles. Find $P(\text{red})$.

a. $\frac{3}{5}$

b. $\frac{8}{17}$

c. $\frac{2}{5}$ ✓

d. $\frac{7}{15}$

$P(\text{red}) = \frac{6}{15} = \frac{2}{5}$.

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